Chapter 4 – Installing Ubuntu Server

Ubuntu Server is the operating system that will host the Wazuh platform. It provides a lightweight, stable, and production-ready environment for running server applications.

Note: This handbook assumes Ubuntu Server is installed before proceeding to Wazuh installation.

4.1 Download Ubuntu Server

Always download Ubuntu Server from the **official Ubuntu website**.

Official Website:

https://releases.ubuntu.com/22.04/

You-tube video:

https://www.youtube.com/watch?v=4C8jLS3Y7hca

Recommended Version:

Ubuntu Server 22.04 LTS (Long Term Support)

Why LTS?

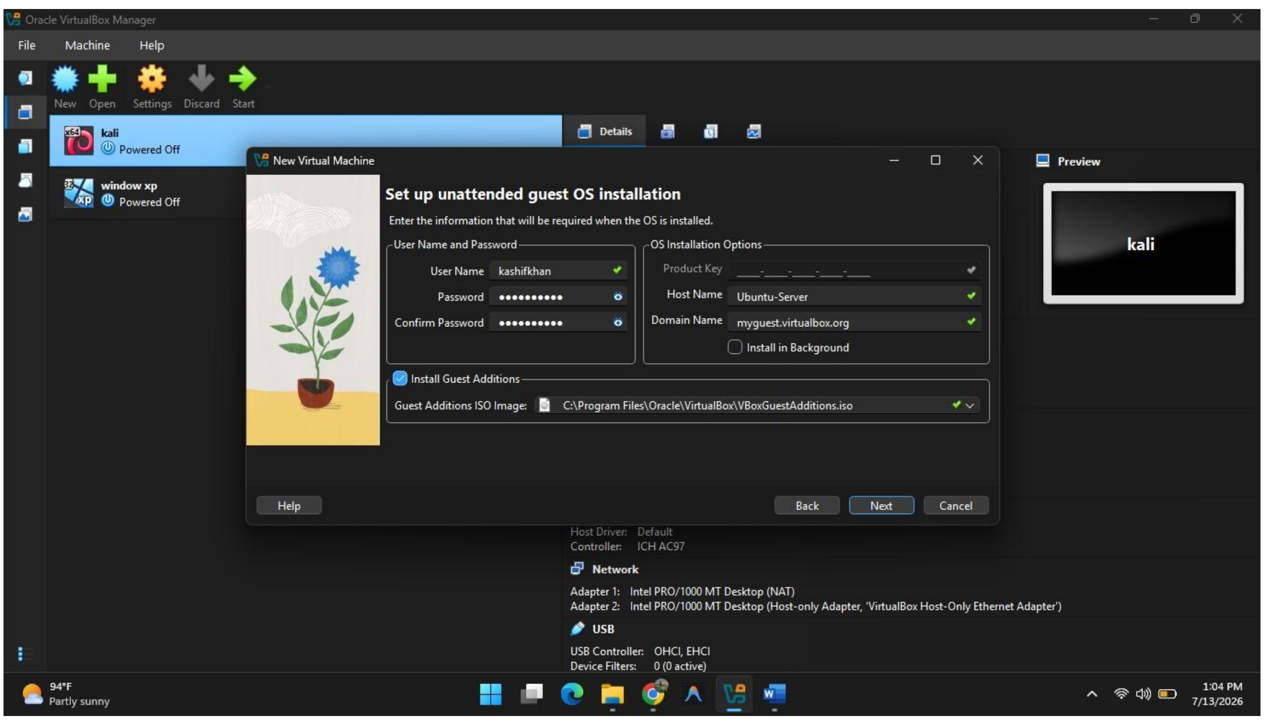
- Stable release
- Long-term security updates
- Recommended by Wazuh
- Commonly used in production environments

4.2 Create the Virtual Machine

Create a new Virtual Machine in Oracle VirtualBox.

Recommended Settings:

Setting	Recommended Value
Name	Ubuntu-Server
Type	Linux
Version	Ubuntu (64-bit)
RAM	6–8 GB (Minimum 4 GB)
CPU	2–4 vCPUs
Disk Size	80–100 GB
Disk Type	VDI (VirtualBox Disk Image)
Storage	Dynamically Allocated



4.3 Virtual Machine Configuration

Processor

Recommended:

- 2–4 CPU Cores

Purpose:

Provides enough processing power for:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

Memory (RAM)

Recommended:

- 8 GB

Minimum:

- 4 GB

Purpose:

Wazuh uses Java services (Indexer), which require sufficient memory for stable performance.

Storage

Recommended:

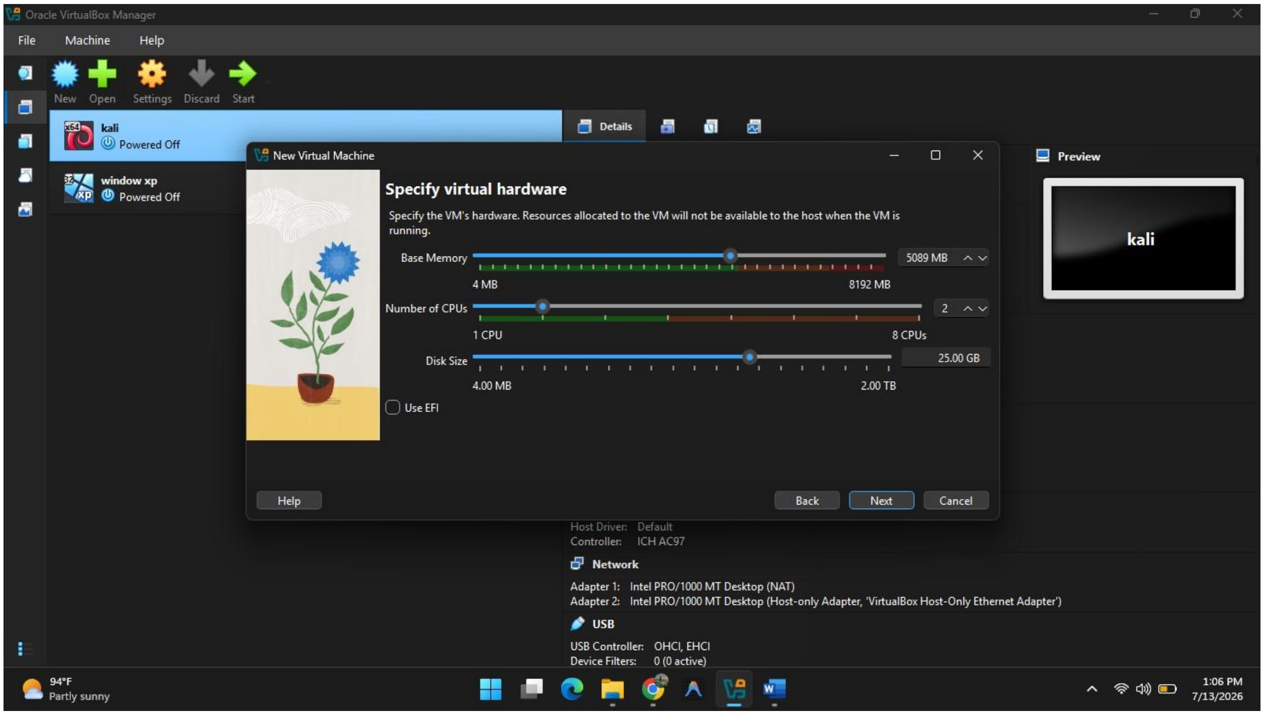
- 80 GB or more

Purpose:

Stores:

- Ubuntu Operating System
- Wazuh Components
- Security Logs
- Alerts
- Index Data

As logs grow over time, adequate storage becomes important.



Network

Recommended:

Adapter 1

NAT

Purpose:

- Internet access
- Software updates
- Downloading packages

Adapter 2

Host-Only Adapter

Purpose:

- Communication with Windows 11
- Communication with Kali Linux
- Wazuh Agent connectivity
- Safe attack simulation

4.4 Installing Ubuntu Server

Boot the VM using the downloaded Ubuntu Server ISO and follow the installation wizard.

During installation:

- Select your preferred language and keyboard layout.
- Configure networking (DHCP is fine initially; we will configure a static IP later).
- Create a hostname (e.g., wazuh-server).
- Create an administrator user and a strong password.
- Install **OpenSSH Server** when prompted.
- Leave optional packages unchecked unless you specifically need them.

4.5 Why Install OpenSSH?

OpenSSH allows secure remote management of the server.

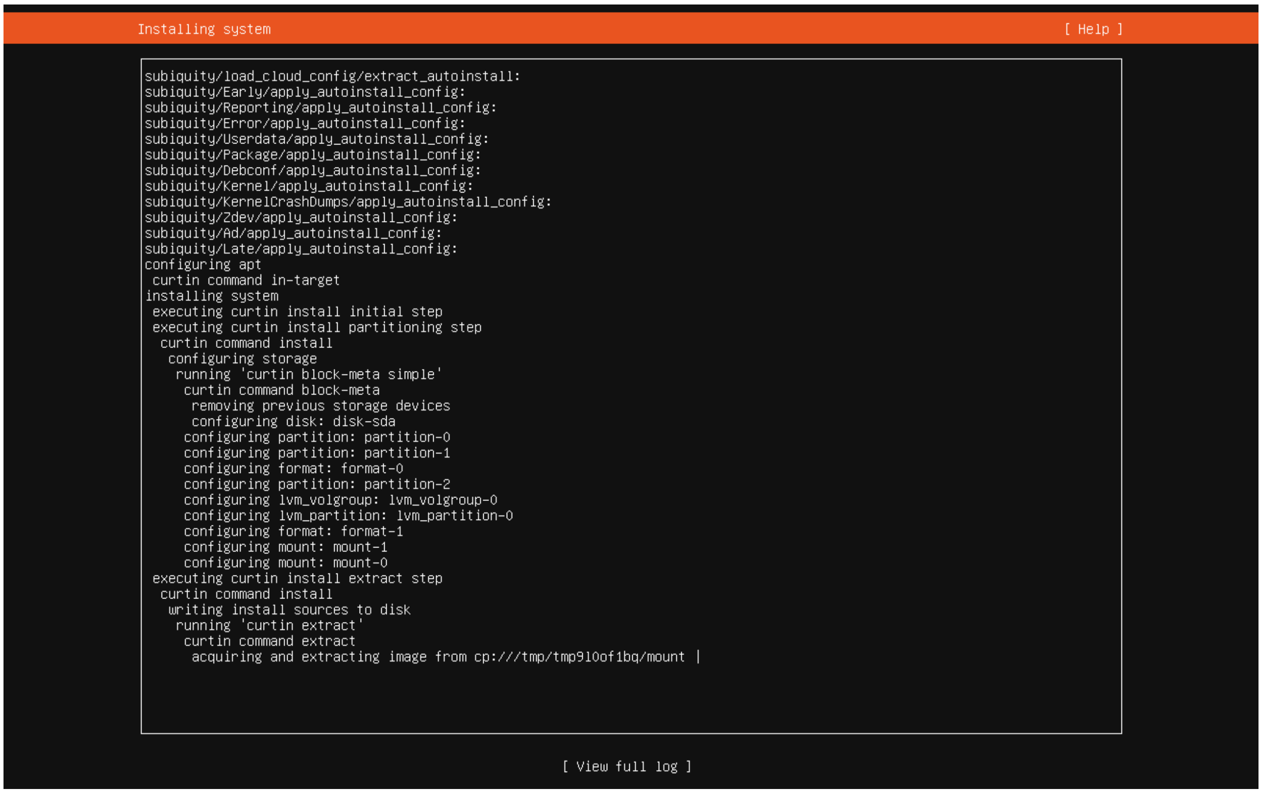
Instead of using the VirtualBox console, you can connect from another machine using SSH.

Example:

ssh username@192.168.56.10

Benefits:

- Remote administration
- Secure encrypted connection
- Common practice in enterprise environments



4.6 First Login

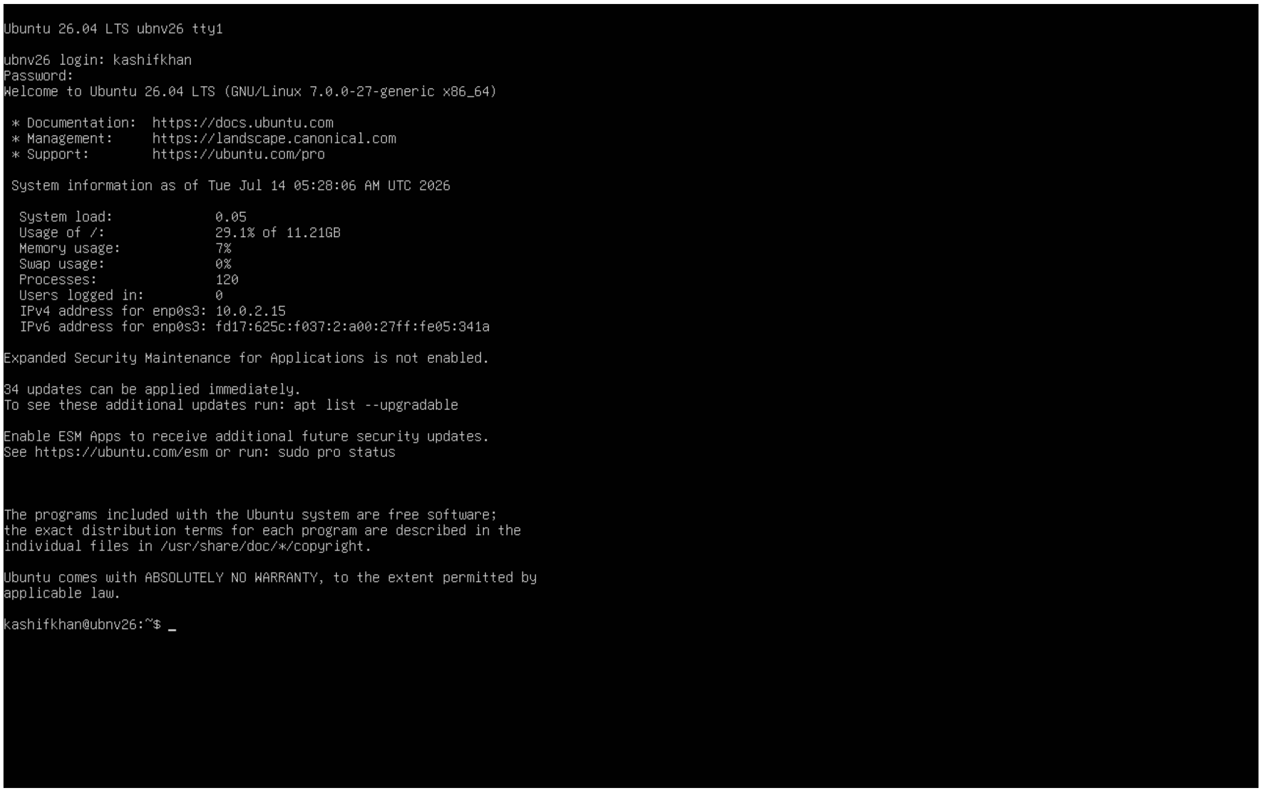
After installation, log in with the username and password created during setup.

Example:

Username: kashif

Password: ********

You should see a Linux terminal prompt, indicating the server is ready for configuration.



4.7 Verify Internet Connectivity

Check that the server can reach the internet before installing packages.

Example:

ping google.com

If replies are received, the server has internet connectivity.

4.8 Verify Ubuntu Version

Run:

lsb_release -a

Purpose:

Displays:

- Ubuntu Version

- Release Number
- Distribution Information

Expected output (example):

Distributor ID: Ubuntu

Description: Ubuntu 24.04 LTS

Release: 24.04

Codename: noble

4.9 Verify Network Configuration

Run:

ip addr

Purpose:

Displays:

- Network Interfaces
- IP Addresses
- Interface Status

This helps verify that both NAT and Host-Only adapters are working correctly.

4.10 Verify Disk Space

Run:

df -h

Purpose:

Displays:

- Disk Usage

- Free Space
- Mounted File Systems

This confirms there is enough storage available for Wazuh.

4.11 Verify System Resources

Run:

free -h

Purpose:

Displays:

- Total RAM
- Used Memory
- Available Memory
- Swap Usage

This helps ensure the VM has enough memory for Wazuh.

Chapter 4 Summary

After completing this chapter, you should have:

- Ubuntu Server 24.04 LTS installed.
- A VirtualBox VM configured with appropriate CPU, RAM, and storage.
- NAT and Host-Only networking configured.
- OpenSSH installed and enabled.
- Internet connectivity verified.
- Basic system information confirmed using Linux commands.

The server is now ready for operating system updates and Wazuh installation.

Commands Learned

Command	Purpose
ping google.com	Test internet connectivity
lsb_release -a	Display Ubuntu version information
ip addr	Show network interfaces and IP addresses
df -h	Display disk usage
free -h	Display memory usage

One-Line Summary

Ubuntu Server provides the stable foundation for the Wazuh platform. Before installing Wazuh, verify the server's network connectivity, operating system version, storage, and memory to ensure it is ready for deployment.